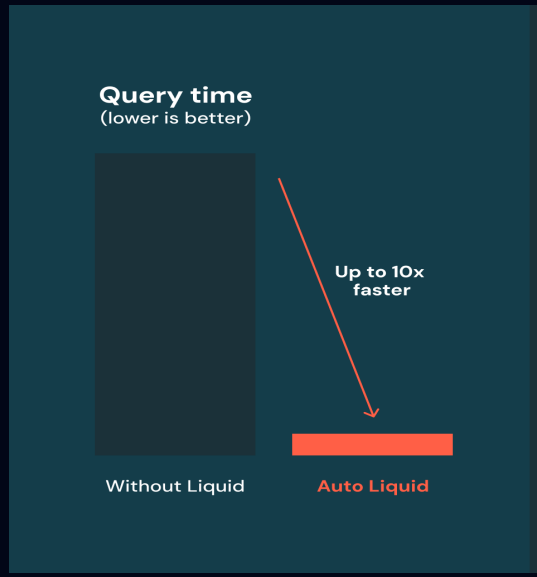
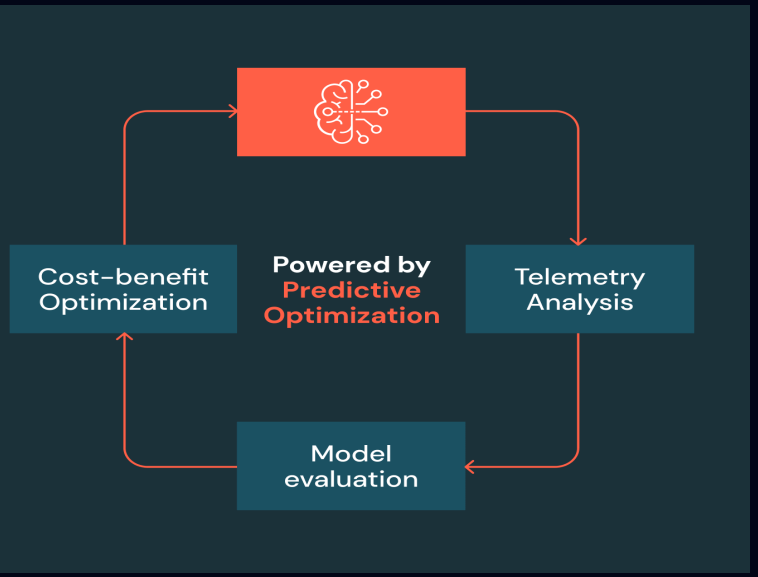
Liquid Clustering

The Smart Way to Organize Your Data





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The Journey

2023 (June) - Databricks introduces Liquid Clustering as a Public Preview feature

2024 (April) - General Availability (GA) released in Databricks Runtime 15.2

2025 (October) - Automatic Liquid Clustering (powered by Predictive Optimization) enters Public Preview

Today - 3000+ active monthly customers writing 200+ PB of data to Liquid-clustered tables

Vision

Replace static Hive partitioning & manual Z-ORDER as default layout strategy

Evolution:

Partitioning → Z-Order → Liquid Clustering

What is Liquid Clustering?

A smart data layout optimization technique that automatically organizes data in your Delta Lake tables

- **Self-tuning:** Adjusts automatically without manual intervention
- **Incremental:** Only rewrites data when necessary, not the entire table
- **Flexible:** Change clustering keys anytime without rewriting existing data
- **Metadata-driven:** Uses Delta Lake transaction logs to track data layout

Why Liquid Clustering?

Old Ways (Partitioning & Z-Order)

- ❌ Manual tuning required
- ❌ Data skew issues
- ❌ Can't change keys without rewrites
- ❌ Complex maintenance overhead

Liquid Clustering

- ✅ Fully automatic
- ✅ Handles skewed data
- ✅ Redefine keys anytime
- ✅ Less operational overhead

How It Works (Technical Magic)

- **Tree-based Algorithm:** Organizes data using a balanced tree structure
- **File-level Clustering:** Groups similar data into files, not just partitions
- **Incremental Process:** New data is clustered on write, old data stays as-is
- **Stateful Tracking:** Delta Log remembers data layout, enabling smart decisions

💡 Think of it like a smart filing system that remembers where everything is and continuously reorganizes for faster access

When to Use Liquid Clustering

- Tables filtered by **high-cardinality columns** (user_id, email, product_id)
- Tables with **skewed data distribution** (uneven data sizes)
- Tables with **rapid growth** that need constant tuning
- Tables requiring **concurrent writes** from multiple sources
- Tables where **query patterns change** frequently over time

🎯 Databricks recommends Liquid Clustering for **ALL new Delta tables**

Performance Wins

Query Speed

2-12x Faster

vs traditional methods

Write Speed

7x Faster

vs partitioning + Z-Order

Creating Liquid Clustered Tables

Simple SQL Syntax:

```
CREATE TABLE orders
CLUSTER BY (customer_id, order_date)
AS SELECT * FROM orders_source;
```

✓ Choose 1-4 clustering columns based on how you filter/join data
✓ That's it! Databricks handles the rest

Monitor clustering status

Simple SQL Syntax:

```
DESCRIBE DETAIL <table name>

-- Shows clustering columns and current optimization level
```

Automatic Liquid Clustering (Advanced)

New in 2025: Let AI choose your clustering keys!

- ✅ Powered by Predictive Optimization
- ✅ Analyzes your query patterns
- ✅ Automatically adapts as patterns change

```
ALTER TABLE my_table
CLUSTER BY AUTO;
```

Key Advantages

- **No More Tuning:** Stop manually optimizing data layouts
- **Flexibility:** Change clustering keys without rewrites
- **Streaming Ready:** Works perfectly with Delta Live Tables
- **Skew-Resistant:** Handles uneven data distribution automatically
- **Cost Savings:** Lower write overhead = reduced infrastructure costs

Real-World Example: E-Commerce

Scenario: Millions of daily orders, queries by customer_id, date range, and region

Old approach: Partition by date, Z-order by customer_id → Complex, high maintenance

Liquid approach: CLUSTER BY (customer_id, order_date) → Done! Handles everything

Result: 10x faster queries, 50% less operational overhead

Getting Started (3 Steps)

- **Identify columns:** What do you filter/join most frequently?
- **Enable clustering:** Add CLUSTER BY clause to CREATE TABLE
- **Optional tuning:** Run OPTIMIZE for existing tables if needed

✅ Use Databricks Runtime 15.2+ for best experience
✅ Works on Unity Catalog managed tables

Key Takeaway

Liquid Clustering = Less work, faster queries, smarter data organization

Recommended for all new Delta Lake tables in Databricks